

Questions with Answer Keys

MathonGo

Q1: 16 March (Shift 1) - Numerical

A certain element crystallises in a bcc lattice of unit cell edge length 270 Å. If the same element under the same conditions crystallises in the fcc lattice, the edge length of the unit cell in Å will be _____. (Round off to the Nearest Integer).

[Assume each lattice point has a single atom]

[Assume $\sqrt{3} = 1.73$, $\sqrt{2} = 1.41$]

Q2: 16 March (Shift 2) - Numerical

Ga (atomic mass 70u) crystallizes in a hexagonal close packed structure. The total number of voids in 0.581 g of Ga is $\times 10^{21}$. (Round off to the Nearest Integer).

Q3: 17 March (Shift 1) - Single Correct

A colloidal system consisting of a gas dispersed in a solid is called a/an:

- (1) solid sol
- (2) gel
- (3) aerosol
- (4) foam

Q4: 18 March (Shift 1) - Single Correct

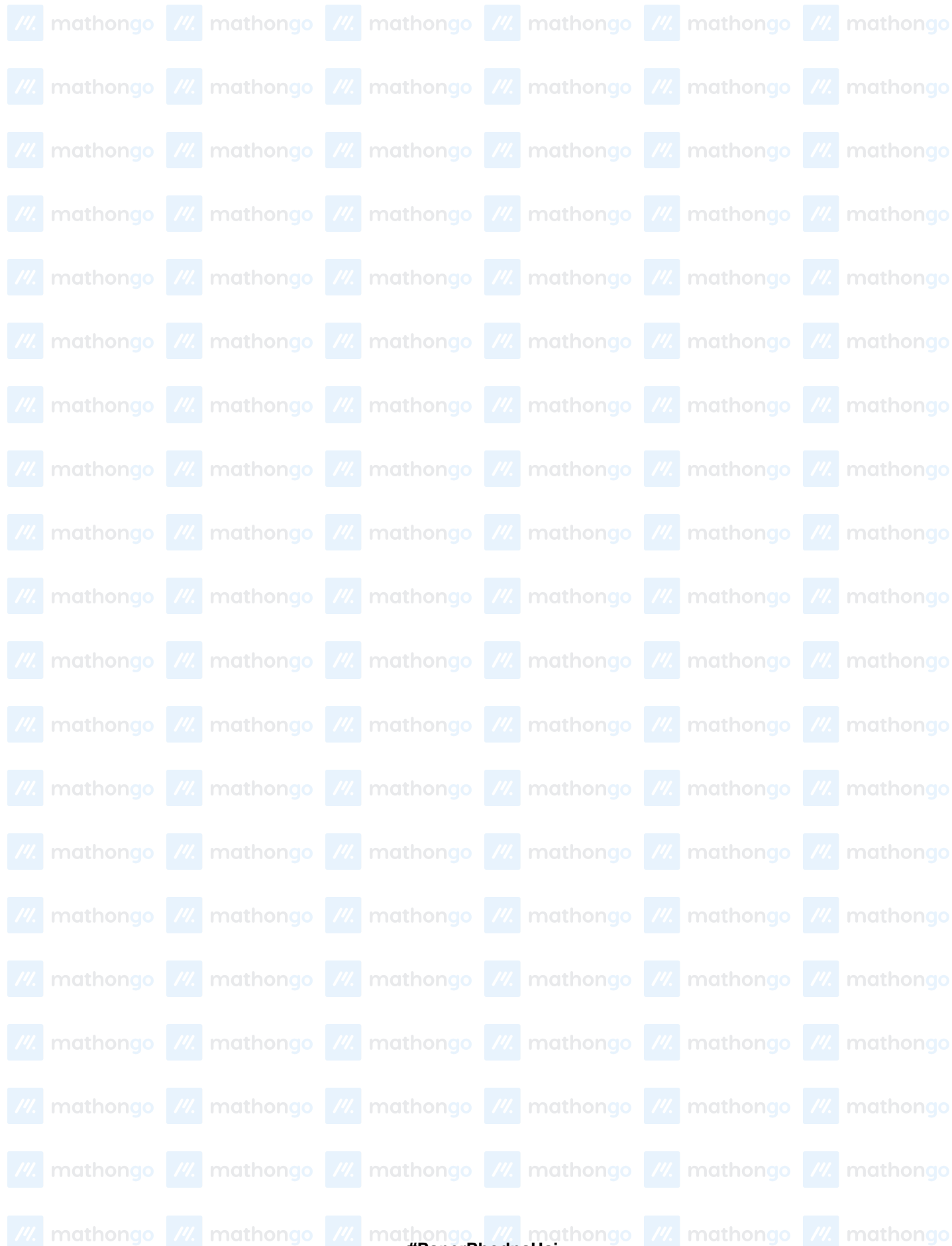
In a binary compound, atoms of element A

form a hcp structure and those of element M

occupy $\frac{2}{3}$ of the tetrahedral voids of the hcp structure. The formula of the binary compound is :

- (1) $M_2 A_3$
- (2) $M_4 A_3$
- (3) $M_4 A$
- (4) MA_3 Official Ans. by NTA (2)

Answer Key**Q1 (33)****Q2 (15)****Q3 (1)****Q4 (2)**



Q1 (33)

For BCC $\sqrt{3}a = 4r$

$$\text{so } r = \frac{\sqrt{3}}{4} \times 27$$

for FCC $a = 2\sqrt{2}r$

$$= 2 \times \sqrt{2} \times \frac{\sqrt{3}}{4} \times 27$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \times 27$$

$$= 33$$

Q2 (15)

HCP structure: Per atom, there will be one octahedral void (OV) and two tetrahedral voids (TV). Therefore

total three voids per atom are present in HCP structure.

→ therefore total no of atoms of Ga will be-

$$= \frac{\text{Mass}}{\text{Molar Mass}} \times N_A = \frac{0.58 \text{ lg}}{70 \text{ g/mol}} \times 6.023 \times 10^{23}$$

→ Now, total Number of voids = $3 \times$ total no. of atoms

$$= 3 \times \frac{0.581}{70} \times 6.023 \times 10^{23} = 14.99 \times 10^{21}$$

$$\simeq 15 \times 10^{21}$$

Q3 (1)

Colloid of gas dispersed in solid is called solid sol.

Q4 (2)

$$M_{12 \times \frac{2}{3}} A_6$$

$$M_8 A_6$$

$$M_4 A_3$$

